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OSUN STATE UNIVERSITY, OSOGBO

NSG 405 — Past Questions & Materials

by Renance

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Institution: Osun State University, Osogbo

Papers in this pack: 4

Levels: 400, 500

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Paper 1 of 4 — Group 16 NSG 405 Presentation Slide New

Level 400 • First Semester • Handout

Osun State University, UNIOSUN,

RotatorCuff

Tears

GROUP 16 NSG 405 PRESENTATION

Rotator Cuff Muscle Injury

The rotator cuff is a group of four muscles and their tendons that surround the shoulder joint. These muscles—the supraspinatus, infraspinatus, teres minor, and subscapularis—work together to stabilize the shoulder joint and enable a wide range of motion.

Imagine your shoulder joint as a ball-and-socket. The rotator cuff is a group of four muscles that surround this joint like a sleeve. They help keep the ball firmly in the socket, allowing for smooth and powerful movement.

A rotator cuff injury occurs when one or more of these muscles or their tendons become damaged. This can range from mild inflammation and tendinitis to a complete tear of the tendon.

Anatomy of Rotator cuff

The rotator cuff is a group of four muscles and their associated tendons that surround the shoulder joint. These structures work together to stabilize the shoulder and allow a wide range of motion. Here's an overview of the anatomy:

01

Supraspinatus

02

Infraspinatus

03 04

Subscapularis

Teres Minor

• Location: Runs • Location: Sits • Location: Lies • Location: Covers

along the top of the shoulder below the supraspinatus below the infraspinatus and the front surface of the

blade (scapula) and attaches scapula (subscapular fossa)

on the back of the runs along the outer edge of

to the top of the humerus and attaches to the front of

scapula. the scapula.

(greater tubercle). the humerus (lesser

• Function: • Function: Assists in

• Function: Initiates tubercle).

shoulder abduction (lifting the Externally rotates the external rotation and

stabilization of the shoulder. • Function:

arm away from the body). shoulder and stabilizes Internally rotates the the joint. shoulder and helps stabilize the joint.

Causes of rotator cuff tears

I. Traumatic Causes

1. Falls: Direct blows to the shoulder or falls onto an outstretched hand can cause rotator cuff tears.
2. Sports injuries: Contact sports, such as football, hockey, or rugby, can lead to rotator cuff tears due to direct blows or sudden stops.
3. Car accidents: Sudden stops or direct impacts can cause rotator cuff tears.

Causes of rotator cuff tears

II. Repetitive Strain Causes

1. Overuse: Repetitive movements, such as throwing, lifting, or overhead reaching, can cause wear and tear on the rotator cuff tendons.
2. Poor posture: Forward head posture or rounded shoulders can put strain on the rotator cuff muscles and tendons.
3. Work-related activities: Jobs requiring repetitive overhead movements, such as construction or painting, can increase the risk of rotator cuff tears.

Causes of rotator cuff tears

III. Degenerative Causes

1. Aging: Rotator cuff tendons naturally degenerate with age, making them more prone to tears.
2. Tendon degeneration: Conditions like tendinosis or tendinopathy can cause degeneration of the rotator cuff tendons.
3. Bone spurs: Bone spurs or osteophytes can rub against the rotator cuff tendons, causing wear and tear.

Causes of rotator cuff tears

V. Risk Factors

1. Age: Increasing age is a significant risk factor for rotator cuff tears.
2. Family history: Having a family history of rotator cuff tears or other shoulder problems increases the risk.
3. Smoking: Smoking can reduce blood flow to the rotator cuff tendons, increasing the risk of tears.
4. Obesity: Excess weight can put additional strain on the rotator cuff muscles and tendons.

Physical Examination

1. Shoulder Inspection: The doctor will visually inspect the shoulder for any signs of swelling, bruising, or muscle wasting.

2. Range of Motion: The doctor will assess the shoulder's range of motion, including flexion, extension, abduction, adduction, rotation, and circumduction.

3. Strength Testing: The doctor will test the strength of the shoulder muscles, including the rotator cuff muscles.

4. Special Tests: The doctor may perform special tests, such as the Neer test, Hawkins test, or empty can test, to assess for rotator cuff impingement or tears.

Physical Examination

1. X-Rays: X-rays may be ordered to rule out other conditions, such as arthritis or bone fractures.

2. Magnetic Resonance Imaging (MRI): An MRI may be ordered to confirm the diagnosis of a rotator cuff tear or to assess the extent of the injury.

3. Ultrasound: Ultrasound may be used to assess the rotator cuff tendons and muscles

Diagnostic Criteria

A diagnosis of a rotator cuff injury is typically made based on a combination of the following criteria:

1. Clinical findings: Abnormal physical examination findings, such as weakness, limited range of motion, or tenderness.
2. Imaging findings: Abnormal imaging findings, such as a rotator cuff tear or tendinosis.
3. Symptom patterns: A consistent pattern of symptoms, such as pain or weakness.

Symptoms of Rotator Cuff Injury

A rotator cuff injury can cause a range of symptoms, including:

Common Symptoms

1. Pain: Aching, burning, or sharp pain in the shoulder, arm, or side of the body.
2. Weakness: Difficulty lifting, rotating, or moving the arm.
3. Limited mobility: Reduced range of motion or stiffness in the shoulder.
4. Snapping or clicking: Audible snapping or clicking sounds when moving the shoulder.
5. Night pain: Pain that worsens at night, disrupting sleep.

Specific Symptoms Based on the Affected Tendon

A rotator cuff injury can cause a range of symptoms, including:

1. Supraspinatus tendon:
 - Pain when lifting the arm or reaching overhead.
 - Weakness when abducting the arm (lifting it away from the body).
1. Infraspinatus tendon:
 - Pain when rotating the arm or lifting it out to the side.

- Weakness when externally rotating the arm (rotating it outward).

Specific Symptoms Based on the Affected Tendon

A rotator cuff injury can cause a range of symptoms, including:

1. Teres minor tendon:

- Pain when lifting the arm or rotating it outward.
- Weakness when adducting the arm (bringing it toward the body).

1. Subscapularis tendon:

- Pain when lifting the arm or rotating it inward.
- Weakness when internally rotating the arm (rotating it inward).

Other Symptoms

1. Muscle atrophy: Shrinking or wasting of the shoulder muscles.
2. Fatigue: Feeling tired or exhausted in the shoulder or arm.
3. Numbness or tingling: Numbness, tingling, or burning sensations in the arm or hand.

Keep in mind that some people may not experience any symptoms at all, or they may attribute their symptoms to other conditions. If you're experiencing persistent or severe shoulder pain, it's essential to consult with a healthcare professional for proper diagnosis and treatment.

Treatment Option

Treatment options for rotator cuff tears depend on the severity of the tear, the patient's age, activity level, and overall health. They are generally divided into non-surgical and surgical approaches:

1. Physical Therapy (PT):

- Strengthening shoulder and arm muscles to improve mobility and stability.
- Specific exercises tailored to the severity of the injury.

1. Medications:

- Nonsteroidal anti-inflammatory drugs (NSAIDs), such as ibuprofen or naproxen, to reduce pain and inflammation.

Treatment Option

1. Corticosteroid Injections:

- Steroid injections may help reduce severe pain, particularly if physical therapy is difficult due to discomfort.

1. Rest and Activity Modification:

- Avoiding overhead activities or heavy lifting that aggravate symptoms.

1. Ice and Heat Application:

- Helps reduce swelling and alleviate pain in acute phases.

Treatment Option

1. Arthroscopic Surgery:

- Minimally invasive; uses a camera and small instruments to repair

the tear.

1. Open Surgery:

- May be necessary for large or complex tears; involves a larger incision for direct repair

1. Tendon Transfer:

- Used if the rotator cuff is irreparable. Another tendon, such as the latissimus dorsi, is transferred to restore function.

Treatment Option

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- Used if the rotator cuff is irreparable. Another tendon, such as the latissimus dorsi, is transferred to restore function.

Surgical Treatments

Surgery is recommended if:

- Symptoms persist after several months of non-surgical treatments.
- There's a complete tear that significantly affects strength or function.
- The patient requires active shoulder use for work or sports.

PREVENTIONS OF ROTATOR CUFF TEARS

I. Maintain Shoulder Flexibility

1. Regular stretching: Engage in regular stretching exercises to maintain flexibility in the shoulder joint.
2. Cross-body stretch: Stretch the shoulder by holding the affected arm with the opposite hand and pulling it across the body.
3. Shoulder rolls: Roll the shoulders forward and backward to loosen the muscles.

PREVENTIONS OF ROTATOR CUFF TEARS

II. Strengthen Surrounding Muscles

1. Rotator cuff exercises: Perform exercises that target the rotator cuff muscles, such as internal rotations, external rotations, and scapular squeezes.
2. Scapular stabilization: Strengthen the scapular stabilizers, including the trapezius, rhomboids, and levator scapulae

muscles.

3. Core strengthening: Engage in core strengthening exercises to improve overall shoulder stability.

PREVENTIONS OF ROTATOR CUFF TEARS

III. Improve Posture

1. Maintain good posture: Avoid slouching or leaning forward, which can put strain on the shoulder muscles.

2. Shoulder blade squeeze: Squeeze the shoulder blades together to maintain proper posture.

3. Chest stretches: Stretch the chest muscles to improve posture and reduce strain on the shoulder muscles.

COMPLICATIONS OF ROTATOR CUFF TEARS

I. Immediate Complications

1. Pain: Severe pain in the shoulder and arm, especially at night.

2. Weakness: Weakness or loss of strength in the shoulder and arm.

3. Limited mobility: Reduced range of motion or stiffness in the shoulder joint.

4. Instability: Feeling of instability or looseness in the shoulder joint.

COMPLICATIONS OF ROTATOR CUFF TEARS

II. Long-term Complications

1. Chronic pain: Persistent pain in the shoulder and arm, even at rest.

2. Muscle atrophy: Shrinking or wasting of the muscles around the shoulder joint.

3. Tendon retraction: Retraction of the torn tendon, making repair more difficult.

4. Bone spurs: Formation of bone spurs or osteophytes, which can cause further irritation.

COMPLICATIONS OF ROTATOR CUFF TEARS

III. Secondary Complications

1. Shoulder impingement: Compression of the rotator cuff tendons between the bones.

2. Bursitis: Inflammation of the fluid-filled sacs (bursae) that cushion the joints.

3. Tendinosis: Degeneration of the tendons, leading to chronic pain and weakness.

4. Osteoarthritis: Wear and tear of the joint cartilage, leading to bone-on-bone contact.

COMPLICATIONS OF ROTATOR CUFF TEARS

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CONCLUSION

Rotator cuff tears are a common condition that can significantly impact a person's quality of life, particularly their ability to perform daily activities and maintain an active lifestyle.

Understanding the anatomy, causes, symptoms, and treatment options is essential for effective management. Early diagnosis and appropriate treatment can lead to better outcomes, reducing the risk of complications and promoting recovery.

Thank you
very much!

Paper 2 of 4 — Group 16 NSG 405 Presentation Slide New

Level 500 • First Semester • Handout

COURSE CODE: NSG 403

TOPIC: Anatomy of the Female Pelvis

6.1 The female bony pelvis

The pelvis is a hard ring of bone, which supports and protects the pelvic organs and the contents of the

abdominal cavity. The muscles of the legs, back and abdomen are attached to the pelvis, and their strength

and power keep the body upright and enable it to bend and twist at the waist, and to walk and run.

The bones of the female pelvis.

The woman's pelvis is adapted for child bearing, and is a wider and flatter shape than the male pelvis. The

pelvis is composed of pairs of bones, which are fused together so tightly that the joints are difficult to see.

Ilium

The major portion of the pelvis is composed of two bones, each called the ilium — one on either side of

the backbone and curving towards the front of the body. When you place your hand on either hip, your

hand rests on the iliac crest, which is the upper border of the ilium on that side. At the front of the iliac

crest, there is a bony protuberance called the anterior superior iliac spine.

Ischium

The ischium is the thick lower part of the pelvis, formed from two fused bones — one on either side.

When a woman is in labour, the descent of the fetal head as it moves down the birth canal is estimated in

relation to the ischial spines, which are inward projections of the ischium on each side. The ischial spines

are smaller and rounder in shape in the woman's pelvis than in that of the man.

Pubic bones and the symphysis pubis

The pubic bones on either side form the front part of the pelvis. The two pubic bones meet in the middle

at the pubic symphysis. (A symphysis is a very strong bony joint.) The pubic symphysis is immediately

below the hair-covered pubic mound that protects the woman's external genitalia.

Sacrum

The sacrum is a tapered, wedge-shaped bone at the back of the pelvis, consisting of five fused vertebrae

(the small bones that make up the spinal column or backbone). At the bottom of the sacrum is a tail-like

bony projection called the coccyx. The upper border of the first vertebra in the sacrum sticks out, and

points towards the front of the body; this protuberance is the sacral promontory — an important landmark for labour and delivery.

6.2 The pelvic canal

The roughly circular space enclosed by the pubic bones at the front, and the ischium on either side at the

back, is called the pelvic canal — the bony passage through which the baby must pass. This canal has a

curved shape because of the difference in size between the anterior and posterior borders of the space

created by the pelvic bones.

The size and shape of the pelvis

The size and shape of the pelvis is important for labour and delivery. Well-built healthy women, who had

a good diet during their childhood growth period, usually have a broad pelvis that is well adapted for

childbirth. It has a round pelvic brim and short, blunt ischial spines. It gives the least difficulty during

childbirth, provided the fetus is a normal size and the birth canal has no abnormal tissue growth causing

an obstruction.

There is considerable variation in pelvis shapes, some of which create problems in labour and delivery. A

narrow pelvis can make it difficult for the baby to pass through the pelvic canal. A deficiency of important

minerals like iodine in the diet during childhood may result in abnormal development of the pelvic bones.

Stunting (being much shorter than average for age) due to malnutrition and/or infectious diseases can also

result in a narrow pelvis.

The pelvic inlet

The pelvic inlet is formed by the pelvic brim. The pelvic brim is rounded, except where the sacral promontory and the ischial spines project into it. The dimensions in centimetres (cm) of the pelvic inlet

are shown in the diagram below.

Diameters of the pelvic inlet, viewed from above.

The pelvic outlet

The pelvic outlet is formed by the lower border of the pubic bones at the front and the lower border of the

sacrum at the back. The ischial spines point into this space on both sides. The diagram below shows the

dimensions of the space that the fetus must pass through as it emerges from the mother's pelvis.

Diameters of the pelvic outlet.

Paper 3 of 4 — GROUP 1 PRESENTATION MEDICAL SURGICAL NURSING (NSG 405)

Level 400 • First Semester • Handout

GROUP 1 PRESENTATION

MEDICAL SURGICAL NURSING (NSG 405)

400 LEVEL

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Introduction to Myasthenia Gravis (MG)

Definition

Myasthenia Gravis (MG) is a chronic autoimmune neuromuscular disorder that causes weakness in the skeletal muscles responsible for movement, breathing, and swallowing. It occurs due to antibodies attacking the acetylcholine receptors at the neuromuscular junction, leading to impaired communication between nerves and muscles. This results in muscle fatigue that worsens with activity and improves with rest.

Epidemiology

MG is a rare disease, affecting 20 to 30 people per 100,000 worldwide.

It can affect anyone, but it is more common in young women (under 40) and older men (over 60).

Neonatal MG can occur when a mother with MG passes antibodies to her baby, though this is usually temporary.

MG can be associated with thymomas (tumors of the thymus gland) in about 10-15% of cases.

Advances in treatment have significantly improved life expectancy and quality of life, making MG a manageable chronic condition rather than a fatal disease.

Pathophysiology of Myasthenia Gravis (MG)

Myasthenia Gravis is an autoimmune neuromuscular disorder that disrupts communication between nerves and muscles, leading to muscle weakness and fatigue.

Step-by-Step Pathophysiology

1. Normal Neuromuscular Transmission

In a healthy neuromuscular junction, nerve impulses (action potentials) travel down motor neurons.

This triggers the release of acetylcholine (ACh), a neurotransmitter, into the synaptic cleft.

Acetylcholine binds to acetylcholine receptors (AChRs) on the muscle cell membrane.

This initiates muscle contraction by generating an action potential in the muscle fiber.

2. Autoimmune Attack on Acetylcholine Receptors

In MG, the body's immune system mistakenly produces antibodies that target the acetylcholine receptors (AChRs) on the muscle surface.

These antibodies block, destroy, or degrade AChRs, preventing acetylcholine from binding effectively.

3. Impaired Neuromuscular Transmission

Due to fewer available receptors, acetylcholine cannot efficiently trigger muscle contractions.

This leads to weaker and fatigued muscles, especially with repeated use.

4. Role of the Thymus Gland

The thymus gland, which helps regulate the immune system, is often abnormal in MG patients.

About 10-15% of MG patients have a thymoma (tumor of the thymus).

The thymus may produce autoantibodies that attack acetylcholine receptors, worsening MG symptoms.

5. Progression and Symptoms

MG symptoms typically begin in the eyes (ptosis, diplopia) and can progress to bulbar muscles (difficulty swallowing, speaking) and limb muscles.

Muscle weakness worsens with activity and improves with rest due to depletion of acetylcholine stores.

6. Myasthenic Crisis

In severe cases, muscle weakness can affect the respiratory muscles, leading to respiratory failure (myasthenic crisis), which requires emergency intervention.

Causes of Myasthenia Gravis (MG)

Myasthenia Gravis is an autoimmune disorder, meaning the body's immune system mistakenly attacks its own cells. The exact cause is not fully understood, but several factors contribute to its development.

1. Autoimmune Attack on Acetylcholine Receptors

The immune system produces autoantibodies that attack acetylcholine receptors (AChRs) at the neuromuscular junction.

This prevents proper transmission of nerve signals to muscles, leading to muscle weakness and fatigue.

2. Role of the Thymus Gland

The thymus gland, which helps regulate the immune system, is often abnormal in MG patients.

Thymoma (tumor of the thymus) occurs in 10-15% of MG cases.

Even in patients without a tumor, the thymus may produce autoantibodies that attack the neuromuscular junction.

3. Genetic and Environmental Factors

MG is not directly inherited, but genetic susceptibility may play a role.

Certain genetic variations in the immune system increase the risk of MG.

Environmental triggers like infections, stress, or surgery might trigger the immune response leading to MG.

4. Other Autoimmune Conditions

MG is associated with other autoimmune diseases, such as:

Rheumatoid arthritis

Lupus

Thyroid disorders (e.g., Hashimoto's thyroiditis, Graves' disease)

5. Drug-Induced MG

Certain medications can worsen or trigger MG symptoms, including:

Beta-blockers (e.g., propranolol)

Calcium channel blockers (e.g., verapamil)

Aminoglycoside antibiotics (e.g., gentamicin)

Neuromuscular blocking agents (used in anesthesia)

Risk Factors for Myasthenia Gravis (MG)

While Myasthenia Gravis can affect anyone, certain factors increase the risk of developing the condition.

1. Age and Gender

MG can occur at any age, but it is more common in:

Women under 40 years old

Men over 60 years old

Late-onset MG (after 50) is often more severe and associated with thymoma.

2. Genetic Susceptibility

MG is not directly inherited, but a family history of autoimmune diseases may increase the risk.

Certain genetic variations in the immune system (e.g., HLA genes) may contribute to MG.

3. Autoimmune Diseases

People with MG are more likely to have or develop other autoimmune disorders, including:

Rheumatoid arthritis

Lupus (SLE)

Thyroid disorders (Graves' disease, Hashimoto's thyroiditis)

4. Thymus Gland Abnormalities

About 10-15% of MG patients have a thymoma (tumor of the thymus gland).

Even without a tumor, the thymus may be enlarged or dysfunctional, triggering autoimmune responses.

5. Certain Medications

Some drugs can trigger or worsen MG symptoms, such as:

Beta-blockers (e.g., propranolol, atenolol)

Calcium channel blockers (e.g., verapamil)

Aminoglycoside antibiotics (e.g., gentamicin, tobramycin)

Neuromuscular blocking agents (used in surgery)

Magnesium-containing medications

6. Viral or Bacterial Infections

Infections may trigger or worsen MG symptoms by activating the immune system.

Some viruses are suspected to play a role in autoimmunity, but no direct link has been proven.

7. Stress and Physical Exhaustion

Emotional and physical stress (e.g., surgery, childbirth, trauma) can exacerbate MG symptoms.

Overexertion can lead to increased muscle fatigue and weakness.

8. Pregnancy and Postpartum Period

Pregnancy can worsen MG symptoms in some women, while others may improve.

Some babies born to mothers with MG may develop transient neonatal MG, which is temporary and resolves with treatment.

Clinical Manifestations of Myasthenia Gravis (MG)

The hallmark of Myasthenia Gravis is fluctuating muscle weakness that worsens with activity and improves with rest. The symptoms vary in severity and may progress over time.

1. Ocular Symptoms (Most Common Early Signs)

Ptosis (drooping eyelids) – One or both eyelids may droop, especially by the end of the day.

Diplopia (double vision) – Due to weakness of the extraocular muscles.

Difficulty focusing – Eyes may not move together properly.

2. Bulbar Symptoms (Affecting Face, Throat, and Speech Muscles)

Dysphagia (difficulty swallowing) – Increased risk of choking or aspiration.

Dysarthria (slurred speech) – Weakness in tongue and facial muscles affects speech clarity.

Weakness in facial muscles – May cause a "mask-like" expression or difficulty in smiling.

Nasal or hoarse voice – Due to weakness in the soft palate and vocal cords.

3. Generalized Muscle Weakness

Neck and limb weakness – Difficulty holding up the head or lifting objects.

Proximal muscle weakness – Affects shoulders, arms, and legs more than distal muscles.

Difficulty in climbing stairs, standing from a seated position, or lifting objects.

Worsening muscle weakness with prolonged activity – Improves with rest.

4. Respiratory Symptoms (Severe Cases, Myasthenic Crisis)

Shortness of breath (dyspnea) – Weakness in respiratory muscles.

Hypoventilation (reduced breathing ability) – Can lead to respiratory failure.

Risk of Myasthenic Crisis – A life-threatening exacerbation requiring ventilatory support.

Pattern of Symptoms

Symptoms typically worsen throughout the day.

Periods of remission and exacerbation may occur.

Trigger factors include stress, infections, fatigue, heat, and certain medications.

Diagnosis of Myasthenia Gravis (MG)

Diagnosing MG involves a combination of clinical evaluation, specialized tests, and imaging studies to confirm the presence of autoantibodies and neuromuscular dysfunction.

1. Clinical Assessment

History and Symptoms

Fluctuating muscle weakness that worsens with activity and improves with rest.

Presence of ocular symptoms (ptosis, diplopia).

Difficulty swallowing, speaking, and breathing in severe cases.

Physical Examination

Fatigue test: Muscle strength weakens with repetitive use.

Peek sign: Inability to keep eyelids closed due to weakness.

Ice Pack Test: Applying ice to the drooping eyelid for 2-5 minutes can temporarily improve ptosis (cold improves neuromuscular transmission).

2. Laboratory Tests

Acetylcholine Receptor Antibody Test (AChR-Ab):

Detects autoantibodies against acetylcholine receptors (positive in 85-90% of generalized MG cases).

Muscle-Specific Kinase (MuSK) Antibody Test:

Used if AChR antibodies are negative (MuSK MG tends to have more bulbar

symptoms).

3. Electrophysiological Tests

Repetitive Nerve Stimulation (RNS):

Measures muscle response to repeated nerve signals (shows decreasing response in MG).

Single-Fiber Electromyography (SFEMG):

More sensitive than RNS, detects abnormalities in nerve-muscle transmission.

4. Edrophonium (Tensilon) Test (Rarely used today due to side effects)

A short-acting drug (edrophonium) temporarily improves muscle strength in MG patients.

Positive test = Rapid improvement in muscle weakness (e.g., ptosis improves within seconds).

5. Imaging Studies

Chest CT or MRI

Checks for thymoma (thymus gland tumor), which is present in 10-15% of MG patients.

6. Pulmonary Function Tests (For Severe Cases)

Used to assess respiratory muscle weakness and the risk of myasthenic crisis.

Treatment of Myasthenia Gravis (MG)

The treatment of MG focuses on improving muscle strength, reducing immune system attack, and managing symptoms. It includes medications, surgical options, and supportive care.

1. Medications

a. Acetylcholinesterase Inhibitors (First-line treatment)

Pyridostigmine (Mestinon) - Increases acetylcholine levels at the neuromuscular junction, improving muscle contraction.

Neostigmine (less commonly used)

Side effects: Excessive salivation, diarrhea, cramping (can be managed with atropine if needed).

b. Corticosteroids (Immunosuppressive Therapy)

Prednisone - Reduces autoimmune attack but may cause weight gain, osteoporosis, and diabetes with long-term use.

c. Immunosuppressants (For Severe or Steroid-Dependent MG)

Azathioprine (Imuran)

Mycophenolate mofetil (CellCept)

Cyclosporine - Used if other immunosuppressants are ineffective.

2. Rapid-Acting Treatments (For Myasthenic Crisis or Severe Cases)

a. Plasmapheresis (Plasma Exchange - PLEX)

Removes circulating autoantibodies from the blood.

Provides temporary relief, often used in myasthenic crisis.

b. Intravenous Immunoglobulin (IVIG)

Provides antibodies to modulate the immune response.

Used in acute exacerbations or when steroids are not tolerated.

3. Surgical Treatment

Thymectomy (Removal of Thymus Gland)

Recommended for patients with thymoma (tumor of the thymus).

May lead to long-term remission in some MG patients, even without thymoma.

4. Supportive and Symptomatic Management

Physical therapy – To prevent muscle atrophy and improve strength.

Speech and swallowing therapy – For patients with bulbar weakness.

Lifestyle modifications:

Avoid stress, infections, excessive heat, and overexertion.

Frequent rest periods to manage fatigue.

Eat soft foods if swallowing is difficult.

5. Myasthenic Crisis Management (Emergency Treatment)

Occurs when respiratory muscles weaken, leading to respiratory failure.

Intubation and mechanical ventilation may be required.

Treated with plasmapheresis or IVIG along with corticosteroids.

Nursing Management of Myasthenia Gravis (MG)

Nursing care for MG focuses on symptom management, preventing complications, patient education, and providing supportive care.

1. Assessment

Neurological & Muscular Assessment:

Monitor muscle strength (especially eyes, face, throat, and limbs).

Assess for ptosis, dysphagia, dysarthria, and fatigue.

Check for respiratory muscle weakness (monitor breathing effort and oxygen levels).

Assess for Myasthenic or Cholinergic Crisis:

Myasthenic Crisis (not enough medication) → Severe muscle weakness, respiratory failure.

Cholinergic Crisis (too much medication) → Muscle twitching, sweating, increased salivation.

Swallowing and Speech Evaluation:

Watch for difficulty swallowing (risk of aspiration).

Monitor for slurred speech and weak voice.

Medication Effectiveness & Side Effects:

Monitor response to pyridostigmine (Mestinon) (check for improved muscle strength).

Watch for side effects of corticosteroids and immunosuppressants.

Nursing Interventions

A. Maintain Airway and Respiratory Function

Assess respiratory rate, oxygen levels, and breath sounds.

Keep suction equipment nearby in case of aspiration.

If respiratory weakness worsens → Prepare for mechanical ventilation (in crisis situations).

B. Prevent Fatigue & Promote Energy Conservation

Encourage rest periods between activities.

Group nursing care to minimize exertion.

Educate on energy-saving techniques (e.g., sitting while dressing).

C. Assist with Eating & Swallowing

Encourage small, frequent meals with soft foods.

Sit upright during meals; avoid thin liquids to reduce aspiration risk.

Keep suction equipment available.

D. Medication Administration

Administer pyridostigmine (Mestinon) on time to maintain muscle function.

Give before meals to improve swallowing ability.

Monitor for side effects of immunosuppressive therapy (infection risk).

E. Prevent Complications

Watch for signs of infection, which can trigger MG exacerbation.

Encourage vaccinations (influenza, pneumonia) to prevent respiratory infections.

Educate on avoiding triggers (stress, infections, extreme temperatures).

3. Patient Education

Medication Adherence: Take medications on time, do not skip doses.

Signs of Crisis: Differentiate between myasthenic vs. cholinergic crisis.

Avoid Triggers: Fatigue, stress, infections, overheating, certain medications.

Medical Alert Bracelet: Wear one to notify healthcare providers of MG.

Emergency Preparedness: Know when to seek immediate medical care (severe weakness, breathing difficulty).

Nursing Diagnoses

1. Ineffective Airway Clearance related to respiratory muscle weakness.

2. Impaired Swallowing related to bulbar muscle weakness.

3. Activity Intolerance related to muscle fatigue.

4. Risk for Aspiration related to dysphagia.

5. Deficient Knowledge related to disease process and self-care.

Summary

Myasthenia gravis (MG) is a chronic autoimmune disorder that causes muscle weakness by disrupting communication between nerves and muscles. It occurs when the immune system produces antibodies that block or destroy acetylcholine receptors, leading to fatigue and weakness, especially in the eyes, face, throat, and limbs. Symptoms often worsen with activity and improve with rest. Treatment includes medications, plasma exchange, or surgery in some cases.

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- or surgery in some cases.

Paper 4 of 4 — MEDICAL SURGICAL NURSING III (NSG 405)

Level 400 • First Semester • Handout

OSUN STATE UNIVERSITY, OSOGBO

FACULTY OF NURSING SCIENCE

COURSE TITLE & CODE: MEDICAL SURGICAL

NURSING III (NSG 405)

GROUP 7

GROUP TOPIC: REACTION TO INTERNAL FIXATION

DEVICES.

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LEVEL: 400LEVEL

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OUTLINE.

1. Overview of Internal Fixation Devices
2. Biocompatibility and Material Reactions
3. Inflammatory and Immune Reactions
4. Infections Related to Internal Fixation Devices
5. Mechanical and Structural Complications
6. Nursing Care and Management
7. Conclusion
8. References

Overview of Internal Fixation Device

Definition: Internal fixation devices are medical implants used to stabilize and join broken bones or fractures, allowing for proper healing and preventing further injury.

Types:

1. Plates: Thin, flat devices used to bridge gaps between bone fragments, providing stability and support.
2. Screws: Threaded devices used to secure plates, rods, or other implants to bone, holding fragments together.

3. Rods: Long, thin devices used to stabilize long bone fractures, such as femur or humerus fractures.
4. Nails: Hollow, tubular devices used to stabilize long bone fractures, often used in combination with screws.
5. Wires: Thin, flexible devices used to stabilize small bone fragments or to provide additional support.
6. Pins: Thin, rigid devices used to stabilize small bone fragments or to provide additional support.

Purpose:

1. Stabilization: Internal fixation devices provide stability to fractured bones, allowing for proper healing.
2. Alignment: Devices help maintain proper bone alignment, ensuring correct healing and minimizing deformity.
3. Support: Devices provide support to weakened or damaged bones, reducing pain and promoting mobility.

Indications for use:

1. Fractures: Internal fixation devices are used to treat various types of fractures, including comminuted, displaced, or unstable fractures.
2. Osteoporotic fractures: Devices are used to stabilize fractures in osteoporotic bones, which are more prone to fracture.
3. Nonunion or malunion: Devices are used to treat fractures that have failed to heal properly (nonunion) or have healed in an incorrect position (malunion).
4. Bone tumors: Internal fixation devices may be used to stabilize bones affected by tumors.
5. Deformity correction: Devices are used to correct deformities, such as scoliosis or clubfoot.

These devices are typically made from metals like stainless steel, titanium, or alloys, and are designed to be biocompatible and corrosion-resistant. The choice of device depends on the type and location of the fracture, as well as the patient's overall health and activity level.

Biocompatibility

Biocompatibility refers to the ability of a material to coexist with living tissues without causing adverse biological responses. Biocompatibility is essential for internal fixation devices, such as implants, screws, and plates, to ensure safe and effective treatment.

Types of Biocompatibility

1. Cytocompatibility: The ability of a material to support cell growth and function.
2. Histocompatibility: The ability of a material to integrate with surrounding

tissues.

3. Systemic biocompatibility: The ability of a material to avoid triggering systemic immune responses.

Material Reactions to Internal Fixation Devices

Internal fixation devices can elicit various biological responses, including:

Inflammatory Response

1. Acute inflammation: Immediate response to tissue damage, characterized by increased blood flow, swelling, and pain.
2. Chronic inflammation: Prolonged response, which can lead to tissue damage, scarring, and implant failure.

Immunological Response

1. Immune cell activation: Activation of immune cells, such as macrophages and T-cells, in response to implant materials.
2. Cytokine release: Release of signaling molecules, such as cytokines, which coordinate the immune response.

Tissue-Implant Interface

1. Osseointegration: Direct contact between bone tissue and implant surface, essential for stable implant fixation.
2. Fibrous encapsulation: Formation of a fibrous tissue layer around the implant, which can lead to implant loosening.

Biomaterials Used in Internal Fixation Devices

1. Metals: Titanium, stainless steel, and cobalt-chromium alloys are commonly used due to their strength, corrosion resistance, and biocompatibility.
2. Polymers: Polymethylmethacrylate (PMMA), polyetheretherketone (PEEK), and ultra-high molecular weight polyethylene (UHMWPE) are used for their biocompatibility, flexibility, and resistance to degradation.
3. Ceramics: Hydroxyapatite, alumina, and zirconia are used for their biocompatibility, hardness, and resistance to corrosion.

Factors Influencing Biocompatibility and Material Reactions

1. Material composition: Chemical composition and surface properties of the material.
2. Surface topography: Roughness, porosity, and texture of the material surface.
3. Sterilization methods: Methods used to sterilize the material, such as gamma radiation, ethylene oxide, or autoclaving.
4. Implant design: Geometry, size, and shape of the implant.

Immune and inflammatory reactions to internal fixation

Inflammatory reaction

Inflammatory reactions to internal fixation are uncommon but can occur due to the body's response to the implanted hardware used in orthopedic procedures to stabilize fractures. These reactions can be classified as either infectious or non-infectious, and their causes and manifestations differ.

1. Non-Infectious Reactions:

Non-infectious inflammatory reactions are typically caused by the body's immune response to the implant material. Common examples include:

A. Hypersensitivity Reactions: Some patients may develop an allergic reaction to metals, such as nickel, cobalt, or chromium, used in internal fixation devices.

Symptoms:

1. Persistent pain, swelling, redness, and limited function around the implant site.
2. Eczematous skin rash or dermatitis (less common).

•Treatment:

1. Removal or replacement of the implant with hypoallergenic materials like titanium or ceramic.

B. Wear Debris-Induced Reactions: Wear and tear or corrosion of the implant can release metal ions or debris, leading to local inflammation.

Symptoms

1. Pain and swelling due to osteolysis or peri-implant inflammation.

Treatment

Revision surgery or switching to alternative materials.

C. Foreign Body Reaction: The body may recognize the implant as a foreign object, resulting in a granulomatous inflammatory response.

Symptoms:

Persistent inflammation around the implant without systemic signs of infection.

Treatment:

Removal of the implant if it compromises function or causes significant symptoms.

2. Infectious Reactions:

Infection around the internal fixation hardware, also known as implant-associated infection, can mimic inflammatory reactions. These infections are usually caused by bacteria like *Staphylococcus aureus* or *Staphylococcus epidermidis*.

Symptoms:

1. Pain, swelling, warmth, and redness at the site.
2. Wound discharge or sinus tract formation.
3. Systemic signs like fever and malaise (in severe cases).

Treatment:

1. Surgical debridement with or without implant removal.
2. Antibiotic therapy tailored to the causative organism.
3. In some cases, staged revision surgery

Immune reaction

An immune reaction to internal fixation refers to the body's response to the implanted hardware (e.g., plates, screws, rods) used to stabilize fractures. This reaction can occur because the body recognizes the implant as a foreign material. While most modern fixation devices are biocompatible, certain factors can trigger immune responses. These reactions may be localized or, in rare cases, systemic.

Types of Immune Reactions to Internal Fixation:

1. Hypersensitivity Reactions

Cause

Allergic response to metals (e.g., nickel, cobalt, chromium) commonly found in orthopedic implants.

Symptoms:

1. Localized pain, swelling, and erythema (redness) around the implant site.
2. Chronic inflammation or joint stiffness without infection.
3. Systemic symptoms like rash or dermatitis (rare).

Treatment:

1. Consideration of hypoallergenic materials (e.g., titanium, ceramic) for implant replacement.
2. Symptom management with anti-inflammatory medications.

2. Foreign Body Reaction

Cause:

The body's immune system identifies the implant as foreign, triggering an inflammatory response.

Symptoms:

1. Chronic pain and swelling around the implant.
2. Formation of fibrous tissue or granulomas.

Treatment:

1. Implant removal if symptoms are severe.
2. Revision surgery with more biocompatible materials.

3. Corrosion or Metal Ion Release

Cause:

Corrosion of the implant or wear and tear releasing metal ions (e.g., nickel, chromium, cobalt).

Symptoms:

1. Persistent local inflammation, pain, or osteolysis.
2. Rare systemic effects like fatigue, cognitive dysfunction, or rash (in cases of systemic metal ion release).

Treatment:

Revision surgery to replace the implant with non-corrosive materials (e.g., titanium).

Infections related to internal fixation devices

Infections related to internal fixation devices, such as plates, screws, rods, and pins, are a significant concern in orthopedic and trauma surgery. These infections can result from various factors and are typically categorized based on their timing and presentation. Here's an overview:

Classification

1. Early Infections (Within 2–4 Weeks):

Often caused by virulent organisms like *Staphylococcus aureus*.

Symptoms: Acute inflammation, erythema, warmth, swelling, and purulent drainage.

Systemic signs like fever may be present.

2. Delayed Infections (4 Weeks to 1 Year):

Caused by less virulent organisms like coagulase-negative staphylococci (*Staphylococcus epidermidis*).

Symptoms: Low-grade inflammation, mild pain, and intermittent drainage.

3. Late Infections (>1 Year):

Often hematogenous in origin.

Symptoms: Sudden onset of pain and swelling over the previously healed surgical site.

Pathophysiology

Biofilm Formation: Bacteria adhere to the device and form a biofilm, protecting them from immune responses and antibiotics, making treatment more challenging.

Contamination Routes:

During surgery (primary contamination).

Hematogenous spread from distant infections.

Secondary contamination from adjacent soft tissue infections.

Risk Factors

1. Patient-Related: Diabetes mellitus. Obesity.

Smoking.

Immune suppression or malnutrition.

2. Procedure-Related:

Prolonged surgery duration.

Contaminated or open fractures.

Improper aseptic technique.

Common Causative Organisms

Staphylococcus aureus (methicillin-sensitive or resistant, MSSA/MRSA).

Coagulase-negative staphylococci.

Gram-negative bacilli (Pseudomonas aeruginosa).

Anaerobes (in mixed infection)

Internal fixation devices (e.g., plates, screws, rods) are highly susceptible to biofilm-associated infections due to their non-biological surfaces, which provide a substrate for bacterial attachment. Biofilm formation complicates treatment, as it confers significant resistance to antibiotics and host immune defense.

Challenges in Treating Biofilm-Associated Infections on Fixation Devices

1. Antibiotic Resistance

2. Immune Evasion

3. Chronicity and Recurrence

4. Device-Dependent Challenges

Internal fixation devices are foreign bodies, which reduce local immunity and create a favorable environment for bacterial adhesion and growth.

Mechanical and Structural Complications

Internal fixation devices, such as screws, plates, rods, and wires, are critical for stabilizing broken bones and aiding proper healing. However, complications can arise due to issues with the device, its impact on surrounding tissues, or its influence on bone healing. These include:

1. **Device Failure:** Devices may loosen, shift (migrate), or break. Loosening can result from weak bones or improper placement, while migration may cause pain or damage to surrounding tissues.

Breakage often occurs under excessive stress, requiring further surgery. Over time, devices can also corrode, leading to inflammation and potential removal.

2. **Impact on Surrounding Tissues:** Devices can irritate nearby soft tissues, press on nerves causing pain or numbness, or lead to infections when bacteria grow on their surfaces. Allergic reactions to materials like nickel in stainless steel may also occur.

3. **Impact on Bone Healing and Remodeling:** Devices can cause stress shielding, where the device absorbs most of the load, weakening the bone over time. Delayed healing or non-union can result from poor stabilization.

Additionally, pressure from the device may lead to bone resorption or impair the natural reshaping of the bone, causing weakened or misshapen structures.

4. Mechanical Overloading and Adjacent Fractures: Premature weight-bearing can overload the device, leading to failure or fractures near the device. Stress concentration around the device can also cause fractures in nearby bones.

While internal fixation devices are essential for fracture management, complications like loosening, migration, breakage, tissue irritation, infection, and delayed healing can occur.

Careful device selection, proper surgical techniques, and close patient monitoring are crucial for minimizing risks and ensuring successful outcomes.

Nursing Care and Management of Patients with Internal Fixation Devices

Assessment:

Vital Signs: Monitor for signs of infection (e.g., fever, tachycardia).

Pain Assessment: Evaluate pain levels using a standardized pain scale and monitor for persistent or worsening pain, which may indicate infection or hardware complications.

Neurovascular Checks: Assess circulation, sensation, and movement distal to the surgical site (e.g., capillary refill, skin color, temperature).

Wound Inspection: Observe for redness, swelling, drainage, or signs of wound dehiscence.

Device Integrity: Palpate for tenderness over the implant site and assess for unusual movement, protrusion, or hardware loosening.

Interventions:

1. Infection Prevention:

Maintain strict aseptic technique during dressing changes.

Administer prescribed antibiotics as ordered.

Encourage proper hand hygiene for both staff and patients.

2. Pain Management:

Administer analgesics as prescribed (opioids or NSAIDs) and monitor effectiveness.

Provide non-pharmacologic pain relief measures (e.g., ice packs, elevation, relaxation techniques).

3. Wound Care:

Perform regular dressing changes using sterile technique.

Monitor drainage and report changes in amount, color, or odor to the provider.

4. Mobility and Rehabilitation:

Encourage early mobilization per provider recommendations to prevent

complications like DVT or stiffness.

Collaborate with physical therapy for range-of-motion exercises and strengthening.

5. Patient Education:

Teach proper wound care and infection prevention at home.

Instruct on signs and symptoms to report, including fever, increased swelling, or persistent pain.

Emphasize weight-bearing restrictions and device protection (e.g., avoiding trauma).

6. Psychosocial Support:

Assess emotional well-being and address anxiety related to the device or recovery process.

Provide resources for coping and rehabilitation programs as needed

Evaluation:

1. Absence of infection (no fever, redness, or drainage).
2. Pain controlled within acceptable levels.
3. Proper wound healing without signs of complications.
4. Demonstrates understanding of discharge instructions and self-care practices.
5. Gradual improvement in mobility and strength.

Conclusion

Internal fixation devices play a crucial role in stabilizing fractures and promoting bone healing, but they also present potential risks, including infections, immune reactions, and mechanical complications. Effective nursing care and management are essential to ensure positive patient outcomes.

By conducting thorough assessments, implementing evidence-based interventions, and providing patient education, nurses can help prevent complications and support recovery. Emphasizing infection prevention, pain management, and early mobilization aids in minimizing risks and improving long-term function.

Ultimately, a collaborative approach involving nurses, physicians, and physical therapists is key to addressing both the physical and emotional needs of patients with internal fixation devices, enabling them to regain mobility and enhance their quality of life.

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